4. Heat Death

1 Weirdly, as of this writing, we are still not actually sure which of these is the main mechanism by which this happens. We just see the star go off and we know at least one white dwarf was involved.

2 As much as the rest of us physicists might find it frustrating to admit, the guy had a lot of pretty good ideas.

3 You can tell you're in a demanding field when just SAVING THE UNIVERSE is not enough.

4 If you were sitting in a different galaxy in a different part of the universe, you would also define your observable universe as being a sphere about 45 billion light-years in radius, centered on your own position. "Observable universe" is a subjective, literally self-centered, concept.

5 As we have seen in Chapter 2, defining "now" is tricky.

6 The relative-size-increase factor of the universe is 1 plus the redshift, so something nearby, at redshift 0, is in a universe whose size is the same as ours.

7 The Hubble radius is not technically a horizon, in the physics sense of the word. The particle horizon is; it's a limit beyond which we cannot possibly obtain information about anything. The Hubble radius is just the radius at which the CURRENT expansion speed is the speed of light, but it changes over time, and, as we've just discussed, objects can cross into it. People sometimes call it a horizon, but many cosmologists will get very worked up if you use that term.

8 Not literally, obviously. That would be both impossible and extremely inadvisable.

9 The other laws are somewhat less exciting, though they do start with zero, just to be weird. Briefly, they are: 0) If one thing is in thermal equilibrium with another thing, and a third thing is in equilibrium with that, they're all in equilibrium with each other. 1) Energy is conserved and perpetual motion machines are